RESEARCH PAPER · NLP / RLHF ALIGNMENT

DiaTool-DPO

Multi-Turn Direct Preference Optimization for Tool-Augmented LLMs

5-State MDP Formulation · Automatic Preference Pair Construction · Near GPT-4o Performance

April 2026

arxiv.org/abs/2504.02882

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

TA-LLMs must control conversation flow across 3 decisions — but SFT on expert trajectories alone provides no negative signal for incorrect dialogue patterns.

Three Failure Modes of SFT-Only Training

Failure Mode 1 — Hallucinated Tool Invocations

Model calls tools with insufficient or missing parameters, fabricating slot values instead of asking the user.

Root cause: no exposure to "incorrect call" negative examples

Failure Mode 2 — Missing Slot-Filling Questions

When a query lacks required arguments, the model skips clarification and proceeds directly to (wrong) tool calls.

Root cause: SFT copies correct paths; never learns to pause and ask

Failure Mode 3 — Failed Out-of-Scope Rejection

When no suitable tool exists, model still attempts a tool call rather than issuing an appropriate rejection response.

Root cause: tool rejection is rare in expert data; model never learns it

Why DPO? — Learning from Contrastive Trajectories

SFT Only

- Trains on expert trajectories (correct paths only)
- No negative examples — no preference signal
- Cannot distinguish correct vs. incorrect dialogue flow

✗ Prone to hallucination and missed rejections

+

DiaTool-DPO (SFT + DPO)

- Pairs chosen (correct) vs. rejected (incorrect) trajectories
- Constructed automatically via 5-state MDP definition
- No human labeling required — scales without annotation cost
- DPO loss directly optimizes for preference over dialogue control

✓ Learns slot-filling AND rejection from contrastive signals

Key Insight:

The MDP formulation allows systematic generation of chosen/rejected pairs for every query type — turning dialogue control into a preference learning problem solvable without manual annotation.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

Each TA-LLM interaction maps to one of 5 internal states. Transitions between states define the dialogue trajectory — the foundation for preference pair construction.

Out-of-scope → Reject

×N repeats

S1 — **Initial** — No history, Conversation start → **S2** — **Tool Selected w/o Complete Slots** — Slot-filling QA here → **S3** — **Tool Selected w/ Complete Slots** — All args determined → **S4** — **Wait for Tool Response** — Awaiting API result → **S5** — **Complete** — Results received, Response generated

S1 — Initial State

Conversation begins. If request is out-of-scope, model returns rejection and stays in S1.

→ Type 3 queries end here

S2 — Incomplete Slots

Tool identified but required parameters are missing. Model asks follow-up questions (×N).

→ Type 2 queries cycle here

S3 — Ready to Call

Tool and all argument values are fully determined. Tool call is executed.

→ Types 1 & 2 pass through

S4 — Awaiting Result

Tool call has been dispatched. Model waits for the external API response to arrive.

→ Transitions to S5 on receipt

S5 — Complete

Tool results received. Model generates final completion response.

→ Dialogue ends

State Trajectories by Query Type:

Type 1 (Complete query)
1 → 3 → 4 → 5 (no slot-filling needed)

Type 2 (Incomplete query)
1 → (2×N) → 3 → 4 → 5 (N slot-fill rounds)

Type 3 (Out-of-scope query)
1 → 1 (reject; self-loop)

Note: These are internal (unobserved) states — not labels in the training data. The MDP structure enables the system to automatically infer which trajectory a dialogue took, enabling automatic chosen/rejected pair construction without human annotation.

Three Query Types Define Distinct State Trajectories

Query type determines the correct dialogue path. Misclassification produces hallucinated calls (Types 1&2) or missed rejections (Type 3).

Query Type	State Trajectory	Description	Example User Query
Type 1 Complete query	**1 → 3 → 4 → 5** No slot-filling needed	All required arguments are present in the initial user utterance.	*"Book a flight to Seoul on Friday at 9am"* (destination + date + time all provided)
Type 2 Incomplete query	**1 → (2×N) → 3 → 4 → 5** N rounds of slot-filling QA	Query lacks required arguments. Model asks follow-up questions until slots filled.	*"Book a flight"* (destination and time missing → ask user)
Type 3 Out-of-scope query	**1 → 1** Self-loop; reject and stay	No suitable tool available for request. Tool call must be rejected.	*"Write me a poem about autumn"* (no poetry tool available → reject)

1 Type 1 — Direct Tool Call

- User provides all required slots upfront
- Model proceeds immediately to tool call
- Shortest trajectory: S1→S3→S4→S5

DPO Lesson:
Prevent model from adding redundant slot-filling questions when unnecessary

Chosen: 1→3→4→5 Rejected: 1→2→3→4→5

2 Type 2 — Slot-Filling Dialogue

- User query is missing required parameters
- Model enters S2, asks questions N times
- Trajectory loops on S2 until all slots filled

DPO Lesson:
Prevent slot hallucination — model must ask, not fabricate missing argument values

Chosen: 1→2×N→3→4→5 Rejected: 1→3→4→5

3 Type 3 — Tool Rejection

- No suitable tool exists for the request
- Model must decline and stay in S1
- Self-loop: S1→S1 (rejection path)

DPO Lesson:
Learn to identify when no tool matches; reject gracefully instead of hallucinating

Chosen: 1→1 Rejected: 1→3→4

Automatic Paired Trajectory Construction Without Human Labels

Each chosen/rejected pair teaches a specific lesson about dialogue control — generated automatically from MDP state definitions, no annotation required.

Query	Chosen Trajectory	Rejected Trajectory	Learning Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2×M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090 / 562	Easy Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2×N)→3→4→5	1→(2×M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	—	1→1	Tool call accept	2,089 / 562	Easy Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2×N)→3→4	Tool call reject	562	Hard

Key: All pairs constructed automatically — no human labeling. Green = Chosen (correct path), Red = Rejected (incorrect path). Source: DiaTool-DPO Table 1 (arxiv.org/abs/2504.02882)

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Evaluated on FunctionChat-Bench. SFT + DiaTool-DPO wins Macro Avg for all 3 model sizes. DPO-only is insufficient — it complements, not replaces, SFT.

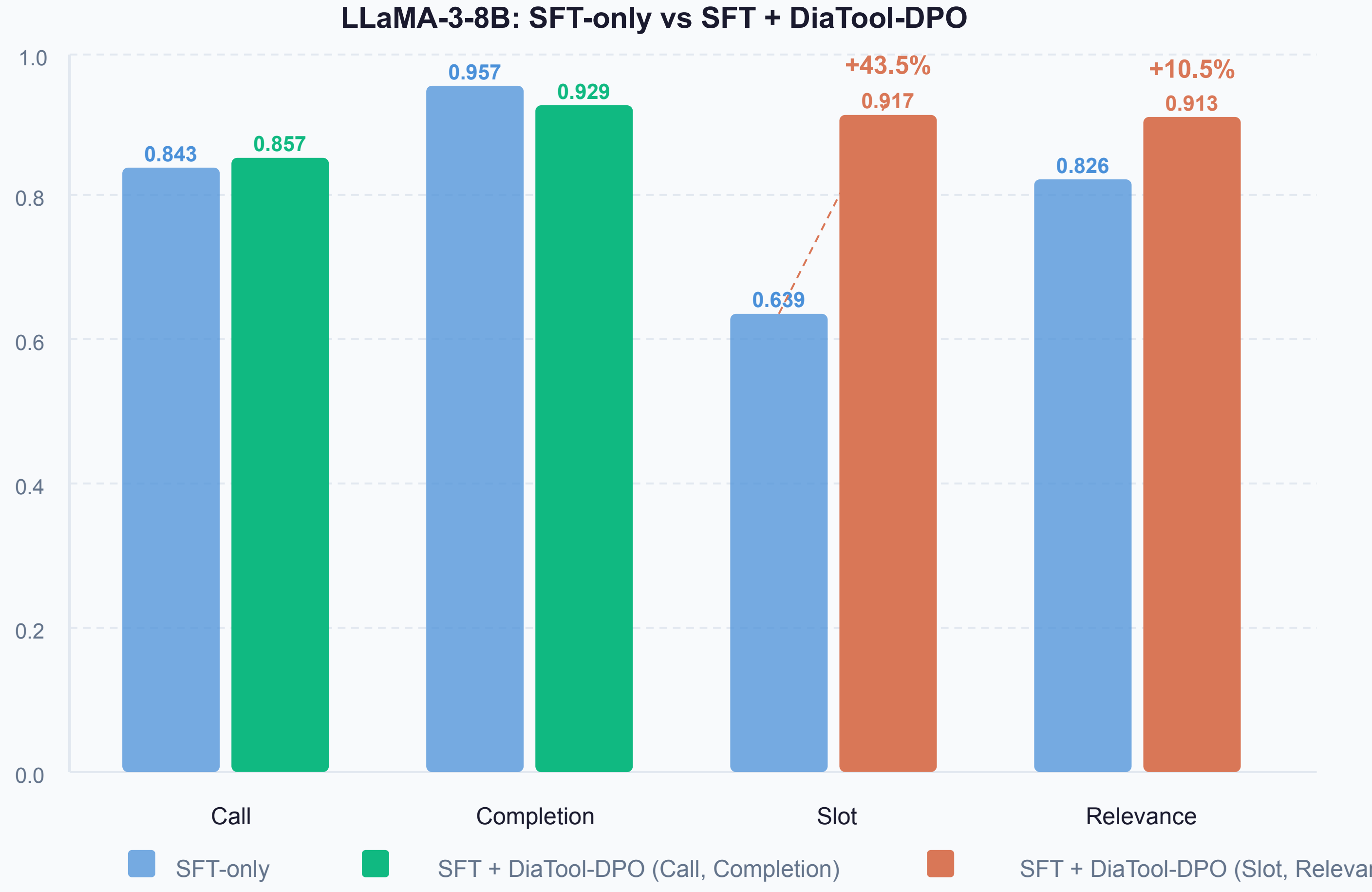
Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B							
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	**SFT + DiaTool-DPO**	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B							
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	**SFT + DiaTool-DPO**	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B							
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	**SFT + DiaTool-DPO**	0.857	0.929	0.917	0.913	0.905	0.904

Best per group   Runner-up best   DPO-only

Source: FunctionChat-Bench evaluation (arxiv.org/abs/2504.02882, Table

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

Adding DiaTool-DPO to SFT produces the most dramatic improvements on LLaMA-3-8B — Slot and Relevance gains far exceed the other model sizes.



Slot Accuracy — Biggest Ga **+43.5%** improvement over SFT-only

0.639 → **0.917**

SFT-only → SFT + DiaTool-DPO

Slot-filling questions now asked correctly; hallucination of missing args eliminated

Relevance — Tool Rejectio **+10.5%**

0.826 → **0.913**

SFT-only → SFT + DiaTool-DPO

Out-of-scope requests now rejected correctly

Full Metric Summary (LLaMA-3-8B)

Micro Avg:	0.844 →	**0.905**	(+7.2%)
Macro Avg:	0.816 →	**0.904**	(+10.8%)

GPT-4o level: 94.8% info gathering, 91% tool rejection

Near parity with GPT-4o on smaller open-source models

Source: arxiv.org/abs/2504.02882

KEY TAKEAWAYS

RL-Based Dialogue Control Opens Path to Production-Ready TA-LLMs

1. MDP Formulation Enables Systematic Pairing

5-state model (Initial → ... → Complete) defines exactly which trajectories are correct or incorrect for each of the 3 query types.

→ Principled preference learning framework

2. Automatic Dataset — No Human Labels

10 pair types across 3 query categories are generated programmatically from MDP definitions. Scales without annotation cost or human effort.

→ Replicable pipeline for any TA-LLM dataset

3. DPO Complements SFT — Not Replaces It

DPO-only performs poorly (LLaMA Slot: 0.056). SFT + DiaTool-DPO achieves best Macro Avg across all three model sizes (3.1B and 8B).

→ Two-stage training is essential

4. Largest Gains in Slot & Rejection

LLaMA-3-8B: Slot +43.5% (0.639→0.917), Relevance +10.5% (0.826→0.913) over SFT-only. Macro Avg reaches 0.904 — near GPT-4o level.

→ 94.8% info gathering, 91% tool rejection

5. Language-Agnostic & Production-Ready

Primary experiments in Korean; method generalizes to English and any TA-LLM deployment context. Opens RL-based dialogue control to open-source

→ Applicable to any production TA-LLM system

DiaTool-DPO: MDP + DPO = Better Tool-Augmented LLM Dialogue

arxiv.org/abs/2504.02882